Syed Talal Wasim

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EDUCATION

University of Bonn — Jan 2024 – Ongoing

Ph.D. in Computer Vision — *Bonn, Germany*

- Supervisor: Professor Juergen Gall | Computer Vision Group | 6 publications at top-tier venues (CVPR, ICCV, ICLR)
- Proposed MixANT (ICCV 2025), the first state-space model with dynamic forget-gate selection via mixture-of-experts, advancing stochastic dense action anticipation past prior state-of-the-art on 50Salads, Breakfast, and Assembly101.
- Co-developed Video-Panda (CVPR 2025), the first encoder-free video-language model, replacing standard 300M to 1.4B parameter visual encoders with a 45M-parameter Spatio-Temporal Alignment Block for a 6.5× parameter reduction and 3 to 4× throughput gain; training scaled across hundreds of GPUs using PyTorch DDP.
- Contributing author on RedSage (ICLR 2026), an open-source 8B cybersecurity LLM trained on a self-curated 11.8B-token continual-pretraining corpus and 266K agentic SFT samples; training scaled across hundreds of GPUs using PyTorch FSDP, surpassing baselines by up to +5.59 points on cybersecurity benchmarks (CTI-Bench, CyberMetric, SECURE) and +5.05 points on the Open LLM Leaderboard.
- Three additional contributions related to efficient visual architectures, state-space model scaling, and domain-specific vision-language models, published at CVPR (x2) and IJCV.

Université de Bordeaux — Sep 2019 – Jun 2021

MS Image Processing and Computer Vision — *Bordeaux, France*

- Funded by the Erasmus Mundus Joint Masters Degree (EMJMD) Scholarship
- Awarded ETH Zurich Robotics Summer School and Symposium 2021 Travel and Costs Grant
- Erasmus mobility: Pázmány Péter Catholic University (Hungary, semester 1) and Universidad Autónoma de Madrid (Spain, semester 2)
- MS Thesis: "Automatic Typography Analysis on Figurative Content" (published in JDMDH, 2023)

Computer Vision Lab, EPFL — Feb 2021 – Jul 2021

MS Thesis Intern — *Lausanne, Switzerland*

- Supervisor: Dr. Mathieu Salzmann
- Investigated automatic typography analysis on figurative content; trained transformer-based image classifiers for serif classification and a vector-based model (DeepSVG) for font similarity learning; released two annotated font datasets and open-source code.

Habib University — Sep 2015 – Jun 2019

BS Electrical Engineering, Minor in Computer Science — *Karachi, Pakistan*

- Graduated with the Dean's Medal (1st position in program) and Best Thesis Award; Dean's and President's Honor List in all 8 semesters
- Merit Scholarship (65% tuition) plus High Academic Achievement Scholarship (additional 10% per semester based on performance)
- BS Thesis: "SquadBot: A Multi-Agent Robotics Teaching and Research Platform"

Stanford University — Jun 2017 – Aug 2017

Summer International Honors Exchange Program — *Stanford, USA*

- Coursework: Technology Entrepreneurship; Leading Trends in IT; Smart Cities & Communities

WORK EXPERIENCE

Mohamed Bin Zayed University of AI — **Apr 2022 – Dec 2023**

Research Associate — *Masdar City, Abu Dhabi, UAE*

- Supervisor: Professor Salman Khan | Computer Vision Department | 6 publications at top-tier venues (CVPR, ICCV, NeurIPS)
- Designed VideoGrounding-DINO (CVPR 2024), the first open-vocabulary spatio-temporal video grounding model, surpassing prior work by 4.88 m_vIoU on HC-STVG and 1.83% accuracy on YouCook-Interactions.
- Designed Video-FocalNets (ICCV 2023), a convolutional alternative to spatio-temporal self-attention for video recognition, replacing transformer attention with focal modulation to match transformer accuracy across five benchmarks (Kinetics-400/600, SS-v2, Diving-48, ActivityNet-1.3) at lower computational cost; subsequently licensed via U.S. patent.
- Led NeurIPS 2023 work on hardware-resilient text-guided image classifiers, using GPT-3-derived text-embedding initialization to deliver a 5.5× average (up to 14×) reliability gain against transient hardware errors with only 0.3% accuracy drop.
- Three additional contributions related to prompt-based foundation model adaptation and efficient long-video segmentation, published at CVPR, ICCV, and WACV.

Empathic Computing Lab, University of South Australia — **Jul 2020 – Mar 2021**

Research Intern — *Remote (COVID-19)*

- Supervisor: Professor Dr. Mark Billinghurst
- Researched multimodal emotion recognition by integrating facial micro-expressions with EEG, GSR, and PPG signals; published in *Frontiers in Psychology* (2022) showing that involuntary micro-expressions are more reliable affect indicators than macro-expressions.

SKILLS

Research Areas: Multimodal Foundation Models, Video Understanding, Vision-Language Models, Efficient Neural Architectures, Long-Form Activity Anticipation, Representation Learning, State-Space Models
Programming: Python (Advanced), Bash / Shell scripting
Deep Learning: PyTorch, HuggingFace Transformers, distributed training (DDP, FSDP, DeepSpeed)
HPC and Tooling: SLURM, Docker, Weights & Biases, OpenCV, NumPy, Pandas, Scikit-Learn
Languages: English (C2, Expert), Urdu (Native)

ACADEMIC SERVICES

- **Conference / Journal Reviewer:** CVPR, ICCV, ECCV, WACV, ACCV, NeurIPS, ICLR, ICML, AAAI, TPAMI, TNNLS, TIP, TMLR, IJCV, Pattern Recognition
- **Outstanding Reviewer at NeurIPS**

AWARDS and GRANTS

- **MENA Winter School on ML 2026:** Best Poster Award (NVIDIA DGX Spark prize); Invited Talk at the winners' announcement covering MixANT (ICCV 2025) and Video-Panda (CVPR 2025)
- **MENA Winter School on ML 2025:** Best Poster Award; Winner, 2025 AI Challenge on Drosophila Neurological Disease Classification; Winter School Grant
- **Compute Award (Jupiter Supercomputer):** 54.0 million GPU compute hours for Holistic Multi-modal Egocentric Video Forecasting
- **Compute Award (Leonardo Supercomputer):** 3.0 million GPU compute hours for Large-Scale Open-Vocabulary Video Understanding and Anticipation
- **Compute Award (Leonardo Supercomputer):** 2.5 million GPU compute hours for Large-Scale Robust Vision Language Models
- **Compute Award (Leonardo Supercomputer):** 1.6 million GPU compute hours for Encoder-Free Video Language Models

Full List of Publications and Patents

CONFERENCES

1. N. Suryanto, M. Naseer, P. Li, **S. T. Wasim**, J. Yi, J. Gall, P. Ceravolo, and E. Damiani, "RedSage: A cybersecurity generalist LLM," in *ICLR*, 2026
2. **S. T. Wasim**, H. Suleman, O. Zatsarynna, M. Naseer, and J. Gall, "MixANT: Observation-dependent memory propagation for stochastic dense action anticipation," in *ICCV*, 2025
3. J. Yi, **S. T. Wasim**, Y. Luo, M. Naseer, and J. Gall, "Video-Panda: Parameter-efficient alignment for encoder-free video-language models," in *CVPR*, 2025
4. A. Shaker, **S. T. Wasim**, S. Khan, J. Gall, and F. Khan, "Groupmamba: Parameter-efficient and accurate group visual state space model," in *CVPR*, 2025
5. D. Velayudhan, A. Ahmed, M. Alansari, N. Gour, A. Behouch, T. Hassan, **S. T. Wasim**, N. Maalej, M. Naseer, J. Gall, M. Bennamoun, E. Damiani, and N. Werghi, "STING-BEE: Towards vision-language model for real-world x-ray baggage security inspection," in *CVPR*, 2025
6. A. Shaker, **S. T. Wasim**, M. Danelljan, S. Khan, M.-H. Yang, and F. Khan, "Efficient video object segmentation via modulated cross-attention memory," in *WACV*, 2025
7. **S. T. Wasim**, M. Naseer, S. Khan, M.-H. Yang, and F. Khan, "VideoGrounding-DINO: Towards open-vocabulary spatio-temporal video grounding," in *CVPR*, 2024
8. **S. T. Wasim**, K. H. Soboka, A. Mahmoud, S. Khan, D. Brooks, and G.-Y. Wei, "Hardware resilience properties of text-guided image classifiers," in *NeurIPS*, 2023
9. **S. T. Wasim**, M. U. Khattak, M. Naseer, S. Khan, M. Shah, and F. Khan, "Video-FocalNets: Spatio-temporal focal modulation for video action recognition," in *ICCV*, 2023
10. M. U. Khattak, **S. T. Wasim**, M. Naseer, S. Khan, M.-H. Yang, and F. S. Khan, "Self-regulating prompts: Foundational model adaptation without forgetting," in *ICCV*, 2023
11. **S. T. Wasim**, M. Naseer, S. Khan, F. Khan, and M. Shah, "Vita-CLIP: Video and text adaptive clip via multimodal prompting," in *CVPR*, 2023

JOURNALS

1. H. Suleman, **S. T. Wasim**, M. Naseer, and J. Gall, "StableMamba: Distillation-free scaling of large ssms for images and videos," *International Journal of Computer Vision (IJCV)*, 2026
2. **S. T. Wasim**, R. Collaud, L. Défayes, N. Henchoz, M. Salzmann, and D. Ribes, "Toward automatic typography analysis: serif classification and font similarities," *Journal of Data Mining in Digital Humanities (JDMDH)*, 2023
3. N. Saffaryazdi, **S. T. Wasim**, K. Dileep, A. F. Nia, S. Nanayakkara, E. Broadbent, and M. Billinghurst, "Using facial micro-expressions in combination with eeg and physiological signals for emotion recognition," *Frontiers in Psychology*, 2022

PATENTS

1. **S. T. Wasim**, M. U. Khattak, M. Naseer, S. Khan, and F. Khan, "System and method for modeling local and global spatio-temporal context in video for video recognition," U.S. Patent 18411928, Jul. 2025
2. M. U. Khattak, **S. T. Wasim**, M. Naseer, S. Khan, and F. Khan, "Method and system for adapting a vision-language machine learning model for image recognition tasks," U.S. Patent 18541666, Jun. 2025